Zeyu Wang

thomaszywang@gmail.com | thomaszywang.github.io

EDUCATION

Sun Yat-sen University, China Sep 2021 - Jun 2024
Master in Electronic and Information Engineering GPA: 3.60/4.0

- Research Interests: Image Processing; Deep Learning; LCD Backlight; Display Technologies
- Scholarship: First-Class Scholarship

Admitted to Master's program via recommendation, exempt from entrance exam

China University of Mining and Technology, China Sep 2017 - Jun 2021
Bachelor in Electronic and Information Engineering GPA: 3.61/4.0

- Scholarship: First-Class Scholarship

PUBLICATIONS

Journals

- **Z. Wang**, G. Zou, Y. Shen, B.R. Yang, Z. Qin. Deep learning-based real-time driving for 3-field sequential color displays with low color breakup and high fidelity. ***Optics Express***, 2023, doi: 10.1364/OE.487198
- **Z. Wang**, G. Zou, Y. Li, C. Ma, Z. Zhang, G. Zha, Y. Shen, B.R. Yang, Z. Qin. Two-field sequential color liquid crystal displays with deep learning-enabled real-time driving. ***Optics Letters***, 2023, doi: 10.1364/OL.501567
- G. Zou, **Z. Wang**, Y. Liu, J. Li, X. Liu, J. Liu, B.R. Yang, Z. Qin. Deep learning-enabled image content-adaptive field sequential color LCDs with mini-LED backlight. ***Optics Express***, 2022, doi: 10.1364/OE.459752

Conferences

- **Z. Wang**, G. Zou, B.R. Yang, Z. Qin. 75-3: Field sequential color displays with simultaneously suppressed color breakup and flicker based on multi-objective optimization. ***SID Symposium Digest of Technical Papers***, San Jose, CA, USA, 2022, doi: 10.1002/sdtp.15670
- **Z. Wang**, G. Zou, Y. Shen, B.R. Yang, Z. Qin. Deep Learning Designed Driving Method for Field Sequential Color Displays. ***Frontiers in Optics***, Rochester, NY, USA, 2022, doi: 10.1364/FIO.2022.JW4B.49
- Z. Qin, **Z. Wang**, Q. Wang, G. Zou, Y. Cheng, Y. Liu, Y. Li, C. Ma, Z. Zhang. High-resolution light field display based on a mini-LED field sequential color micro-LCD. ***SPIE Photonics West***, San Francisco, CA, USA, 2024, doi: 10.1117/12.2692767 **(Invited talk)**

Patents

- Z. Qin, G. Zou, **Z. Wang**, H. Liang, B.R. Yang. Adaptive Refresh Rate Field Sequential Color Display Driving Method, Apparatus and Related Device. Chinese Invention Patent (Granted), CN202210380314.3, 2022.

Textbooks

- **Wearable Optoelectronic Display Technology**, edited by Bo-Ru Yang, Science Press, Beijing, China, 2022.
  Contributor to Chapter 2 Photometry, Colorimetry, and Display Performance Evaluation, with the Chapter authored by Zong Qin

INDUSTRIAL PROJECTS

Field Sequential Color Display for Pancake VR System Jan 2023 - Dec 2023
Primary Contributor | In collaboration with China Star Optoelectronics Technology (CSOT)

- Developed a driving method integrating multi-objective optimization and deep learning to enable real-time high-resolution image generation with minimal distortion and color breakup
- Applied the driving method to a Pancake VR prototype featuring dual 2K displays on an FPGA platform, achieving real-time performance at 60 FPS
- CSOT demonstrated the VR prototype at CES 2024, Las Vegas, NV, USA

Driving Method for Field Sequential Color 4K TV Displays Dec 2023 - June 2024

Primary Contributor | In collaboration with Hisense

- Developed a deep learning-based driving method to suppress color breakup, distortion, and flicker in a 85-inch 4K mini-LED TV
- Proposed a two-field driving method that synchronously generates backlight and LC signals using dual lightweight neural networks, reducing refresh rate requirements and data transmission bandwidth
- Validated performance using objective metrics and subjective visual evaluation on large-scale displays

RESEARCH PROJECTS

Driving Method Development for LCoS-based AR-HUD System Jun 2021 - Feb 2022

Primary Contributor | Advisor: Prof. Zong Qin

- Developed an ultra-low-complexity driving method for the system, leveraging global dimming to suppress color breakup while preserving image quality
- Enabled real-time generation of high-resolution images with high brightness and optical efficiency for automotive AR-HUD applications

WORK EXPERIENCE

Tencent, Shenzhen, China Jun 2024 - Nov 2025

Semiconductor supply chain management

- Managed self-developed server components and developed supply plans aligned with business requirements and market conditions
- Coordinated cross-functional teams to improve supply chain efficiency and reduce delays through proactive coordination and data-driven planning
- Participated in supplier evaluation and onboarding for in-house products, and handled production scheduling and lower-tier material management

SKILLS & INTERESTS

Skills: Python, C++, OpenCV, Matlab, Lighttools
Languages: Mandarin, English (IELTS 7)
Interests: Photography, Traveling, Hiking